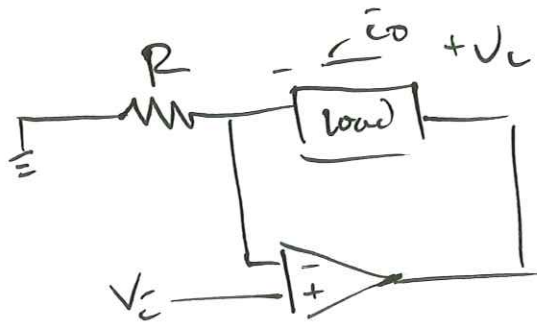


V-I frequency Response



non-f dependent:

$$i_o = A_{vI} V_i - \frac{V_L}{R_o}$$

$$i_o = \frac{a}{R(1+a)} V_i - \frac{1}{R(1+a)} V_L$$

$$a(jf) = \frac{a_o}{1+jf/f_b}$$

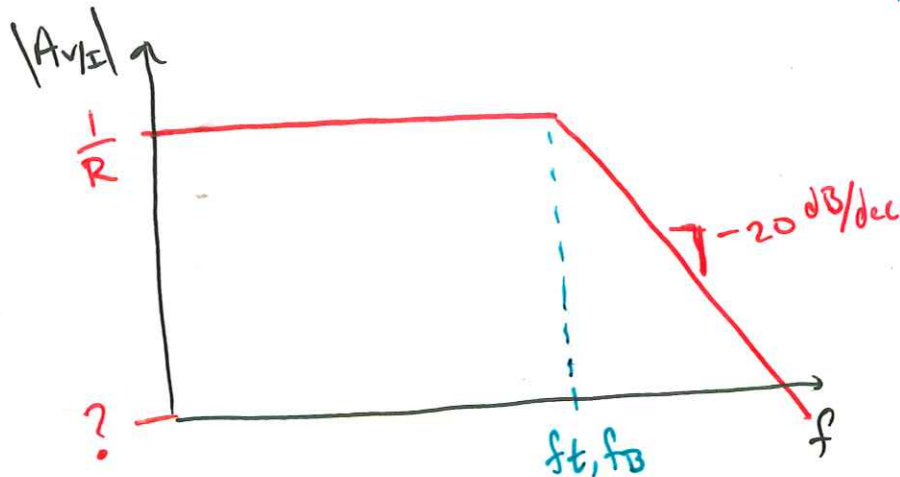
f-dependent

$$A_{vI}(jf) = \frac{a(jf)}{R(1+a(jf))} = \frac{a_o / (1+jf/f_b)}{R(1+a_o / (1+jf/f_b))}$$

$$= \frac{a_o}{R(1+a_o)} \frac{1}{1+jf/f_b(1+a_o)}$$

$$A_{vI}(f) \big|_{f \rightarrow 0} \approx \frac{1}{R}$$

$$f_B = f_b(1+a_o) \approx f_b a_o = f_t$$

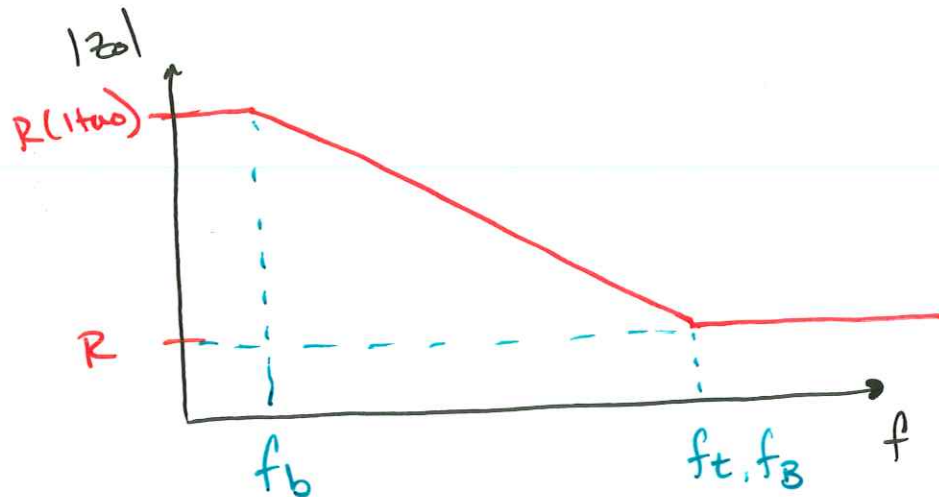


$$R_0 = R(1+a)$$

$$\rightarrow Z_o(jf) = R(1 + a_0 / (1 + jf/f_b)) = R(1 + a_0) \frac{1 + jf/f_c}{1 + jf/f_b}$$

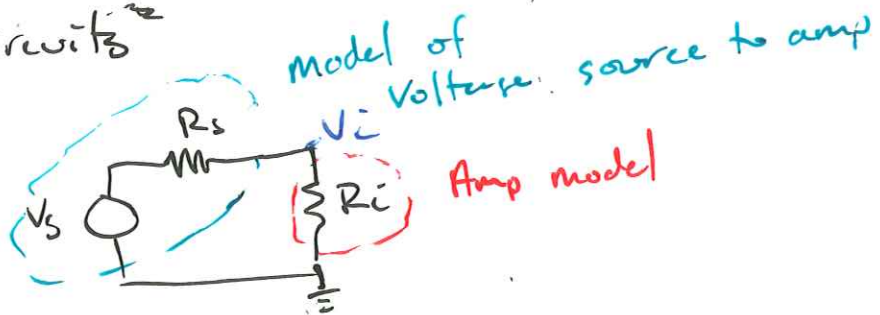
$$f \rightarrow 0 \quad Z_o \approx R(1 + a_0)$$

$$f \rightarrow \infty \quad Z_o \approx R(1 + a_0) \frac{f_b}{f_t} = R(1 + a_0) \frac{f_b}{f_b(1 + a_0)} = R$$



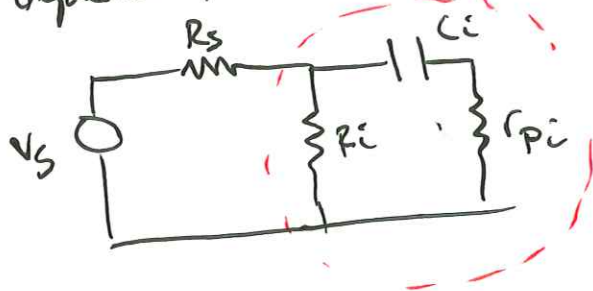
"Resistance" Model of Amp Circuits

for example R_i :



Frequency dependent version

Z_i :



ex.

$$Z_i(jf) = r_o(1 + a_o\beta) \frac{1 + jf/f_B}{1 + jf/f_o}$$

$f \rightarrow 0 \quad Z_i = r_o(1 + a_o\beta) = R_i$

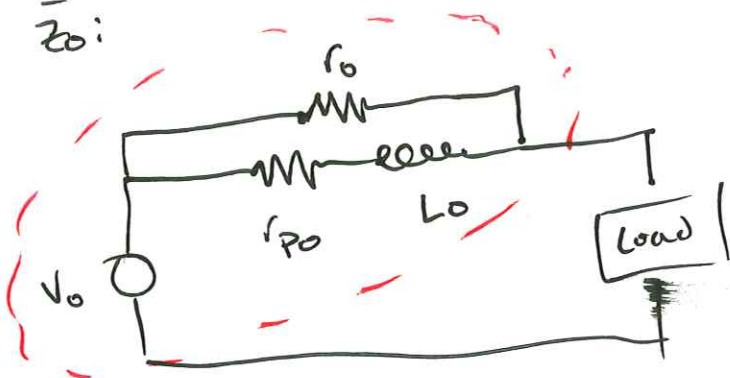
$f \rightarrow \infty \quad Z_i = r_o$

select r_o such that

$$r_o = r_{pi} \parallel R_i$$

select C_i to place f_o, f_B

Z_o :



ex.

$$Z_o(jf) = \frac{r_o}{1 + a_o\beta} \frac{1 + jf/f_B}{1 + jf/f_o}$$

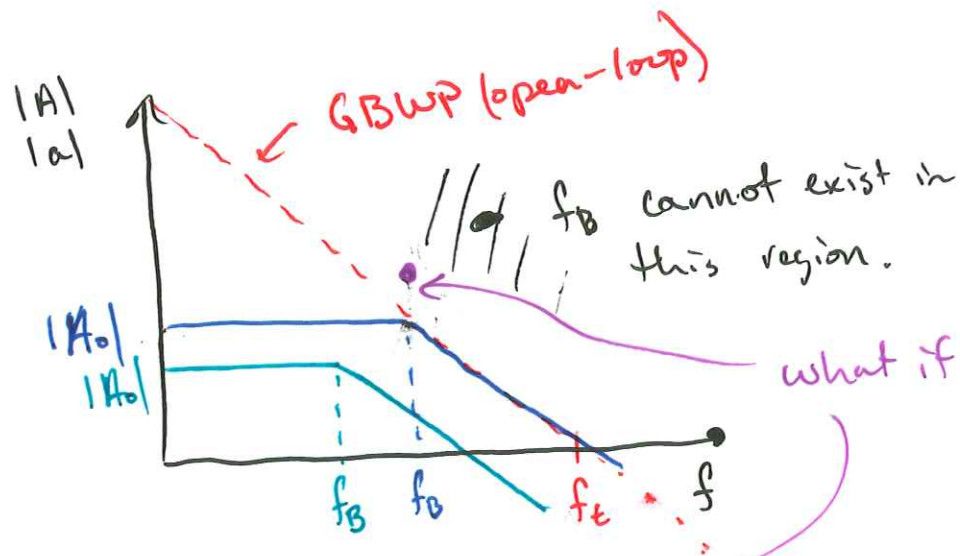
$f \rightarrow 0 \quad Z_o = R_o = \frac{r_o}{1 + a_o\beta}$

$f \rightarrow \infty \quad Z_o = r_o$

$$R_o = r_{po} \parallel r_o$$

select L_o to set f_o, f_B

Bandwidth Considerations



Can only put f_0 on the open-loop GBWP line or underneath.

2 solutions:

- 1) Get a different Op Amp! (need: higher GBWP)
- 2) Cascade Amplifiers:



$$A(jf) = A_1(jf) A_2(jf)$$



$$|A(jf)| = |A_1(jf)| |A_2(jf)|$$

$$\angle A(jf) = \angle A_1(jf) + \angle A_2(jf)$$

Cascaded Amplifiers!

$$A_1(jf) = A_{1,0} / (1 + jf/f_{B1}) \quad A_2(jf) = A_{2,0} / (1 + jf/f_{B2})$$

$$A(jf) = A_1(jf) A_2(jf) = A_{1,0} A_{2,0} / ((1 + jf/f_{B1})(1 + jf/f_{B2}))$$

← 2-pole response,
Decay at -40dB/dec.

if $f_{B1} \neq f_{B2}$ are "close" (within order of magnitude)
 f_B (cascaded bandwidth) relies on both frequencies/poles.

to calculate:

$$|A(jf_{-3dB})| = -3dB = 0.707... = \frac{1}{\sqrt{2}} \quad \leftarrow -3dB \text{ point (bandwidth)}$$
$$= A_{1,0} A_{2,0} / |1 + jf/f_{B1}| |1 + jf/f_{B2}| = \frac{A_{1,0} A_{2,0}}{\sqrt{2}}$$

f_{-3dB}

$$\rightarrow \sqrt{2} = |1 + jf/f_{B1}| |1 + jf/f_{B2}|$$

$$\text{if } f_{B1} = f_{B2} \rightarrow \sqrt{2} = |1 + jf/f_{B1}|^2 = \sqrt{1^2 + \frac{f_{-3dB}^2}{f_{B1}^2}}$$

$$\rightarrow f_{-3dB} = f_B = \sqrt{2^{1/2} - 1} f_{B1}$$

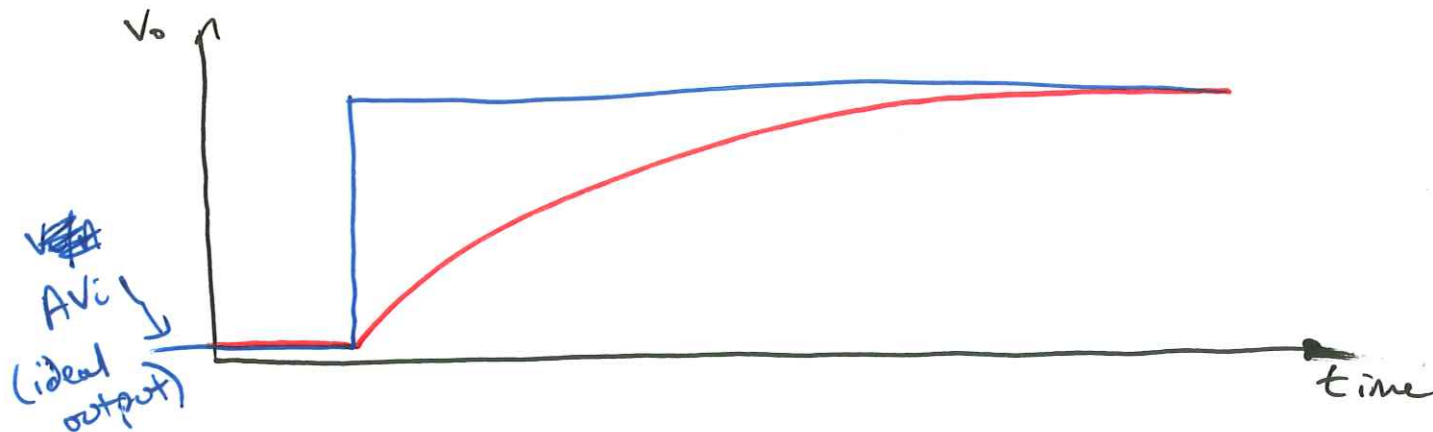
if using N cascaded amplifiers...

$$f_{-3dB} = \sqrt{2^{1/N} - 1} f_{B1}$$

if $f_{B1} \neq f_{B2}$
NOT close,
lower valued one
would be $f_{-3dB} = f_B$!

Transient Responses

$$A(j\omega) = A_0 / (1 + j\omega/\omega_B) \quad \leftarrow \text{1st order system}$$



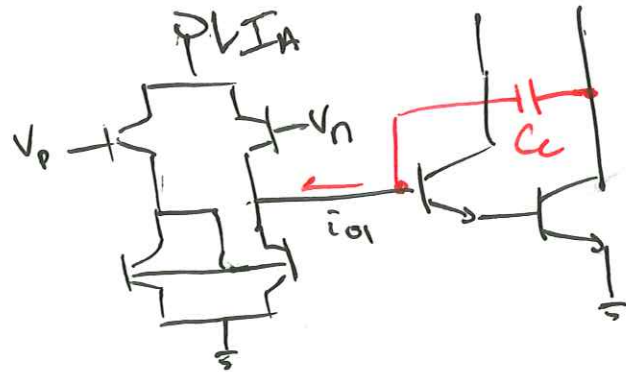
$$\tau = 1/\omega_B = 1/2\pi f_B$$
$$t_r = \frac{2.197}{\omega_B} = \frac{0.35}{f_B}$$

Op-Amp typically fast such that

$$\tau \approx ns$$

"Small Signal Response"
(Linear!)

Large Signal Response...



I_A has a maximum limit due to design. (or a fixed value)

$$\therefore i_{O1} \leq I_A$$

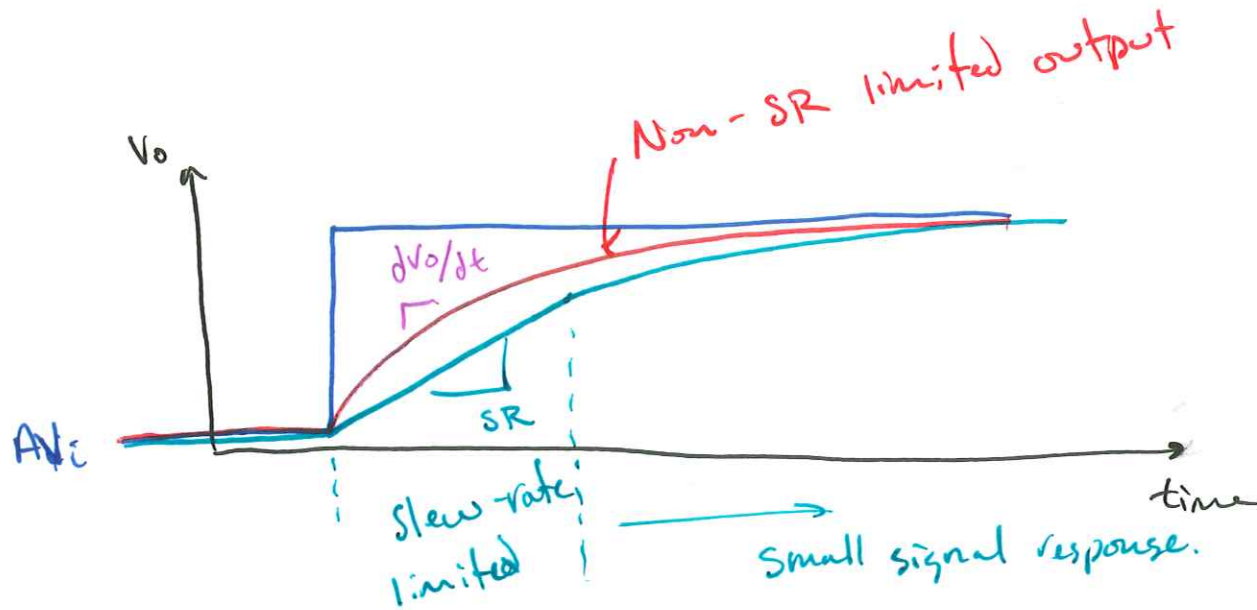
Charging speed of ~~the~~ C_c is limited by this!

$$i_{O1} = C_c \frac{dV_c}{dt} \rightarrow \frac{dV_c}{dt} = \frac{i_{O1}}{C_c} \rightarrow \left. \frac{dV_c}{dt} \right|_{i_{O1}=I_A} = \frac{I_A}{C_c} = \text{SR}$$

"Slew Rate"

Slew Rate: fastest rate at which V_o can change.

$$\text{SR is max } \frac{dV_o}{dt}$$



Non-Linear!

with freq input: to avoid SR-limiting,
want $f \leq \frac{SR}{2\pi V_{out}}$ OR $f \leq \frac{SR}{2\pi V_{max}}$

this f known as the
"full-power-bandwidth" FPB
max. f without SR limiting.

Warning. f_{PB} could be $< f_B$

↑ if ~~max~~ output
stays below V_{sat} .
(V_{max} : ~~the~~ Amplitude of Sine wave)

usable bandwidth lower
of FPB & f_B .

Cascade Example:

Assume amp with $\text{GBWP} = 1\text{MHz}$ ($f_t = 1\text{MHz}$)

want high-gain Audio Amplifier with $f_{-3\text{dB}} \geq 20\text{kHz}$

One Amp: (non-inv. Amp)

$$\text{GBWP} = f_t \approx |A_o| f_B \rightarrow f_B = f_{-3\text{dB}} = 20\text{kHz}$$

$$\rightarrow A_o = 50\text{V/V (max)} = 33\text{dB}$$

two Amps: (identical)

$$f_{-3\text{dB}} = 20\text{kHz} = \sqrt{2^{1/2} - 1} f_{B1} \rightarrow f_{B1} = 31.076\text{kHz}$$

$$\rightarrow A_{o1} = f_t / f_{B1} = 32.18\text{V/V} = 30.15\text{dB}$$

$$A_o = A_{o1}^2 = 1036\text{V/V} = 60.3\text{dB}$$

three Amps: (identical)

$$f_{-3\text{dB}} = 20\text{kHz} = \sqrt{2^{1/3} - 1} f_{B1} \rightarrow f_{B1} = 39.231\text{kHz}$$

$$A_{o1} = 25.49\text{V/V} = 28.13\text{dB}$$

$$A_o = A_{o1}^3 = 16564\text{V/V} = 84.39\text{dB}$$